

GATE

Computer Science & IT



Compiler Design



Syntax Directed Translation

Lecture No. 01



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Recap of Previous Lecture



Topic

Parsers

Topic

Topic

Topics to be Covered



Topic

Parsec

Topic

Topic

→ To Construct LR(0)/SLR(1) we have to create Canonical Collection of LR(0) items.

To create Canonical Collection of LR(0) items,
we use two functions closure() and goto().

Closure(I):-

I is a set of LR(0) items like $A \rightarrow \alpha \cdot B \beta$, $C \rightarrow a$.
 The function closure takes as i/p a set of LR(0) items
 and produces a set of LR(0) items.

Rules to find closure(I):-

1. Initially add every item from I/p to o/p.
2. If $A \rightarrow \alpha \cdot \underline{B} \beta$ is in I, where 'B' is a non terminal and $B \rightarrow \gamma$ is the rule for B, then add $B \rightarrow \gamma$ to closure(I). Repeat this for every newly added item

$$S' \rightarrow S$$

$$S \rightarrow aSa/ab$$

$$\text{closure}(S' \rightarrow \cdot S)$$



$$S \rightarrow \underline{\cdot S}$$

$$S \rightarrow \cdot aSa/ab$$

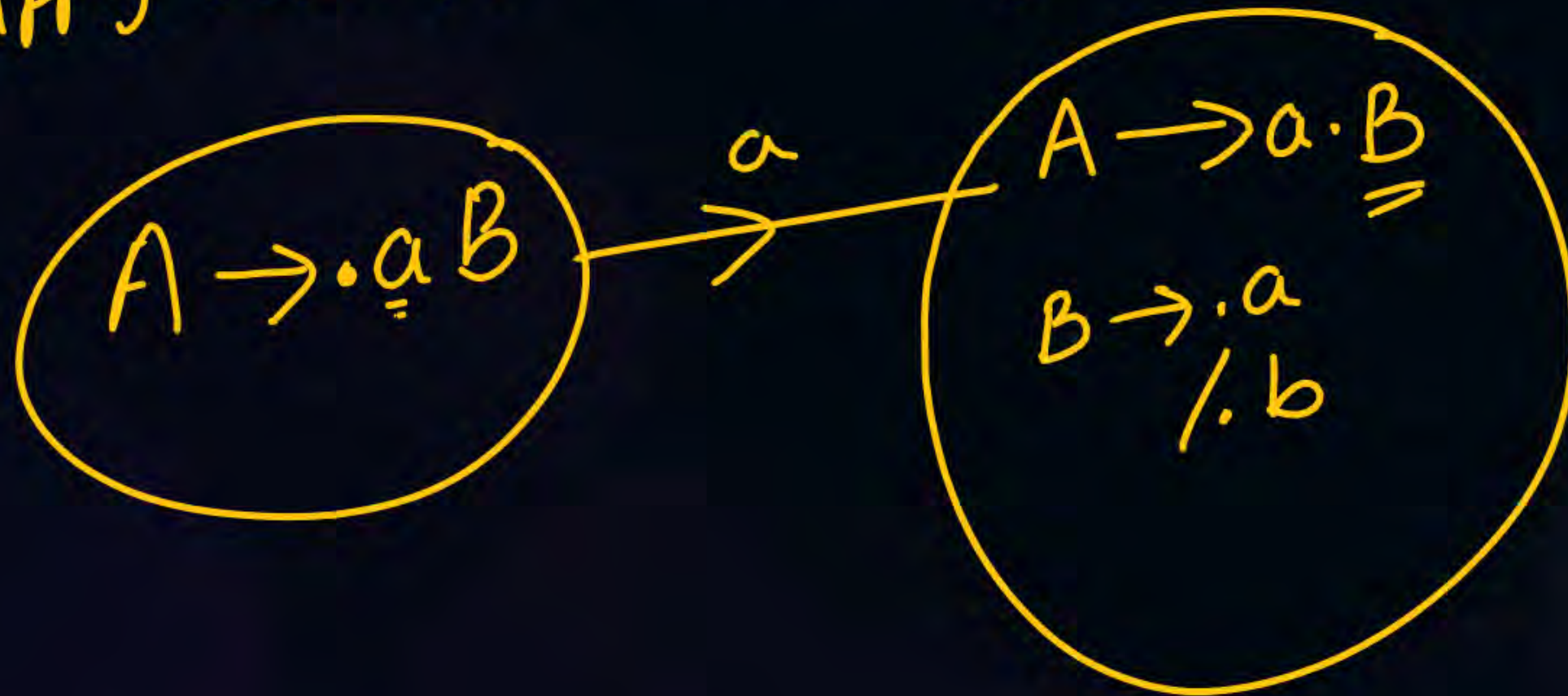
Goto(I, x) :-

where I is a set of LR(0) items and x is a grammar symbol

$$\text{goto}(I, x) = \text{closure} \left\{ A \rightarrow \alpha x \beta / A \rightarrow \alpha \cdot x \beta \text{ is in } I \right\}$$

Goto involves two steps:-

1. Find transition
2. Apply closure () on resulting grammar



Creating Canonical Collection of LR(0) items :-

1. The initial item in C is I_0

$I_0 = \text{closure}(\text{augmented production with } \cdot \text{ at the beginning})$

$$I_0 = \text{closure}(\underline{\underline{S' \rightarrow \cdot S}})$$

2. For each I_i in C and each grammar symbol X in ' G '

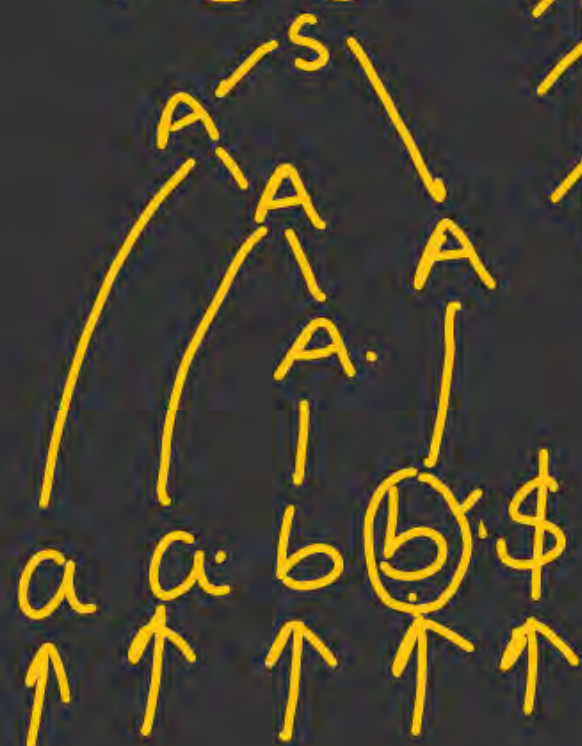
Repeat
while $\text{goto}(I, X)$ is not in set C

and not a empty set-

add $\text{goto}(I, X)$ to ' C '

until no more new set of items
can be added to set C

$S \rightarrow AA$ ①
 $A \rightarrow aA$ ②
 $/b$ ③



$$S \rightarrow |AA| = 2 \times 2 = 4$$

1
S
5
A
2
A
0

acc up =

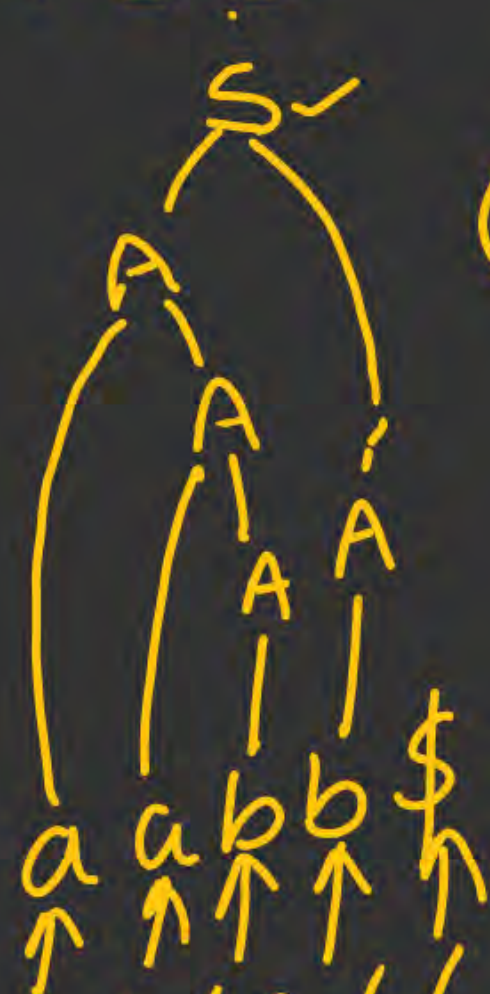
4
b
2
A
6
A
6
A
4
b
3
a
3
a
0

3 min

LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0	S ₃	S ₄	acc	2	1
1				5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	r ₁	r ₁	r ₁		
6	r ₂	r ₂	r ₂		

$S \rightarrow AA$ ①
 $A \rightarrow aA$ ②
 $A \rightarrow b$ ③



accept

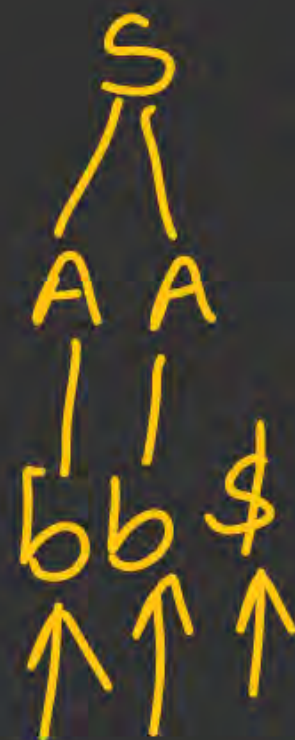
$\checkmark S \rightarrow |AA| = 2 \times 2$
 $\checkmark A \rightarrow |aA| = 2 \times 2$
 $\checkmark A \rightarrow |b| = 1 \times 2$

LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0	S ₃	S ₄	acc	2	1
1				5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	r ₁	r ₁	r ₁		
6	r ₂	r ₂	r ₂		

0	a	3	a	3	b	4	A	6	A	6	A	6	A	2	b	4	A	5	S	1	
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	--

$S \rightarrow AA$ ①
 $A \rightarrow aA$ ②
 $/b$ ③



acc

$A \rightarrow |b| = 1 \times 2$

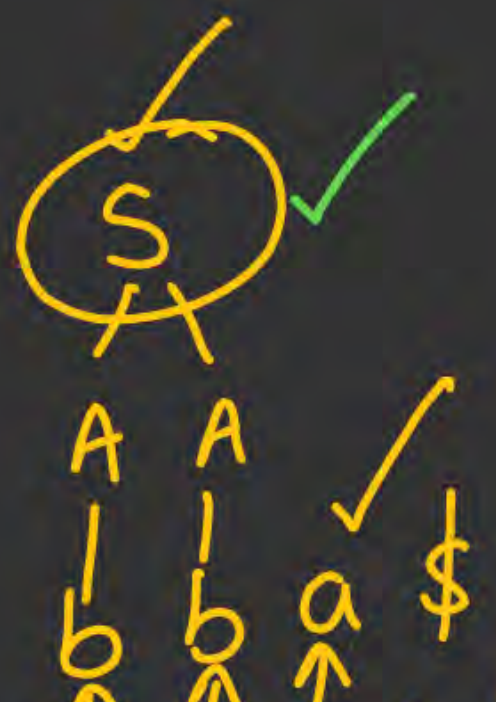
$S \rightarrow |AA| = 2 \times 2 = 4$

	1
S	
5	
A	
4	
b	
2	
a	
4	
b	
0.	

LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0.	S ₃	S ₄	acc	<u>2</u>	1
1				5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	r ₁	r ₁	r ₁		
6	r ₂	r ₂	<u>r₂</u>		

$S \rightarrow AA$ ①
 $A \rightarrow aA$ ②
 $\quad \quad b$ ③



0	b	a	A	a	b	a	A	\$
---	---	---	---	---	---	---	---	----

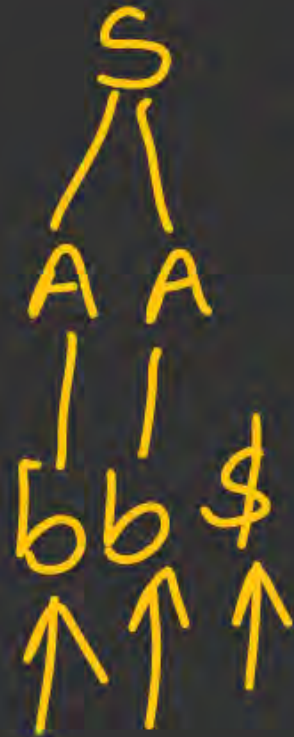
LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0.	S ₃	S ₄	acc	2	1
1	⊙			5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	r ₁	r ₁	r ₁		
6	r ₂	r ₂	r ₂		

	a	b
6	S	R

SLR(1)

$S \rightarrow AA$ ① $fol(S) = \{\$ \}$
 $A \rightarrow aA$ ② $fol(A) = \{a, b, \$ \}$
 $\Rightarrow /b$ ③



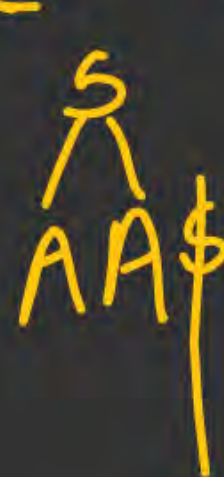
acc

$A \rightarrow |b| = 1 \times 2$
 $S \rightarrow |AA| = 2 \times 2 = 4$

1
S
5
A
4
b
2
A
4
b
0.

LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0.	S ₃	S ₄	acc	2	1
1				5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	✓	✓	r ₁		
6	r ₂	r ₂	r ₂		



4 → r₃

SLR(1)

$S \rightarrow AA$ ①
 $A \rightarrow aA$ ②
 $A \rightarrow b$ ③

$AA \checkmark$
 $\uparrow \uparrow \uparrow$
 $bba\$$

0	1	2	A	2	3	4	A	5	
---	--------------	--------------	---	---	--------------	--------------	---	---	--

LR(0) parsing table

	action			Goto	
	a	b	\$	A	S
0	S ₃	S ₄	acc	2	1
1				5	
2	S ₃	S ₄		6	
3	S ₃	S ₄			
4	r ₃	r ₃	r ₃		
5	r ₃	r ₁	r ₁		
6	r ₂	r ₂	r ₂		

$\begin{matrix} S \\ \wedge \\ AA\$ \end{matrix}$
 $\begin{matrix} S^x \\ \wedge \\ AAa \end{matrix}$
 $\begin{matrix} S \\ \wedge \\ AAb \end{matrix}$
 $4 \rightarrow r_3$

Advantages of LR(0) parser -

- It is very simple to construct LR(0) compared to other parsers
- Each row in the table defines unique action, that is, either shift Δ reduce Δ accept

LR parsing alg 0:- (LR(0), SLR(1), LALR(1), CLR(1))



Let 'x' be the state on top of the stack and 'a' be the symbol pointed by "ip". Action part of parsing table is used to take parsing decision as follows:

1. If action[x, a] = S_i then (Shift action) push 'a'

then state 'i' onto stack. Advance the i/p.

2. If action[x, a] is r_i, then (reduce action). Take ith production

let ith production be A → α. Pop 2*|α| symbols from stack.
and push 'A' on to stack. If 'i' is state below 'A' then
Push GOTO[i, A] onto the stack

3. If action $[x, a] = \underline{\text{accept}}$, then Successful Completion.



SLR(1):-

Reduce actions are placed only in the follow of LHS
 shift and goto are same for LR(0) and SLR(1)
 δ , is placed under '\$'

($S \rightarrow AA$)



SLR(1)

↓
lookahead

SLR(1) parsing table:-

In case of shift moves and goto moves, LR(0) and SLR(1) are same. But only difference is reduce action. Reduce moves are placed only in the foll(LHS). It means it uses one lookahead.

SLR(1) is more powerful than LR(0) because the reduce moves are less in SLR(1) compared to LR(0). Blank entries are made in SLR(1), so end entries are made in SLR(1) compared to LR(0).

So SLR(1) detects errors much faster

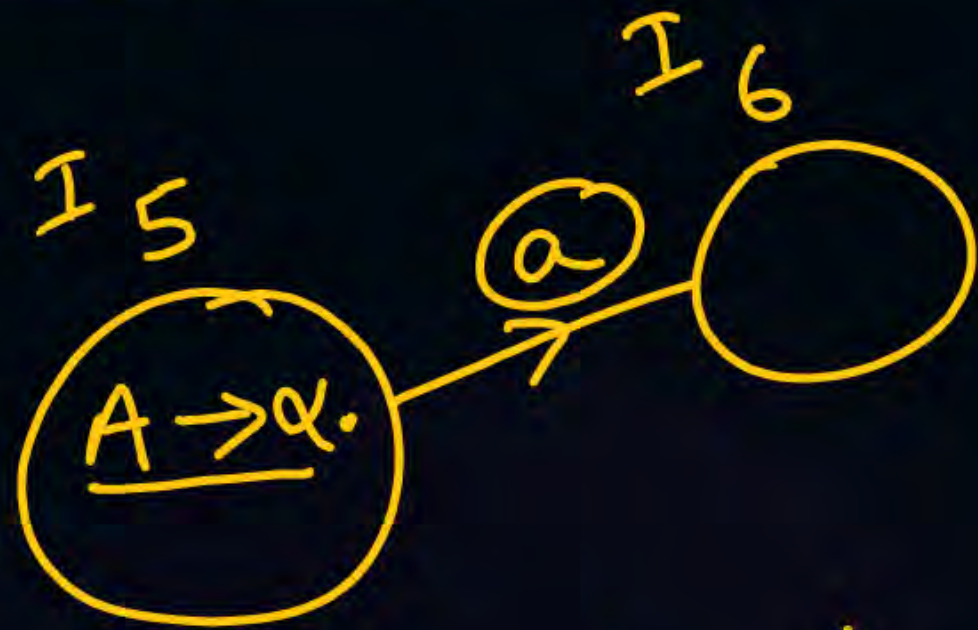


Disadvantages of LR(0) parser:-

- The LR(0) parser can be used to parse a small class of grammars.
- Look ahead is not used in LR(0) in making parsing decisions.

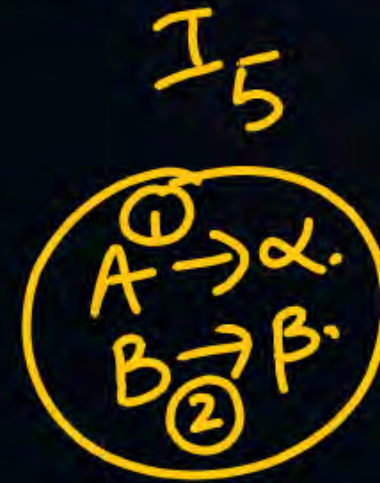
LR(0)
↓
zero
ahead

Testing whether a grammar is LR(0) or not :-



	a	b	\$
5	γ/s	γ	γ

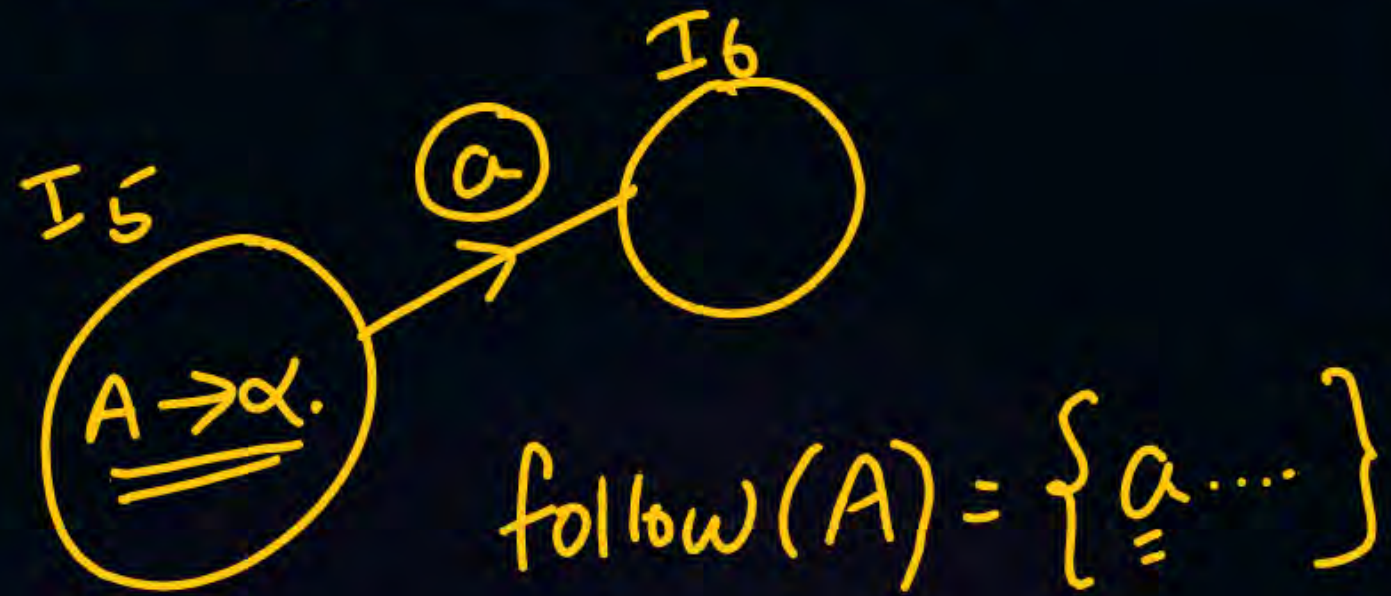
SR-Conflict



	a	b	\$
5	γ_1/γ_2	γ_1/γ_2	γ_1/γ_2

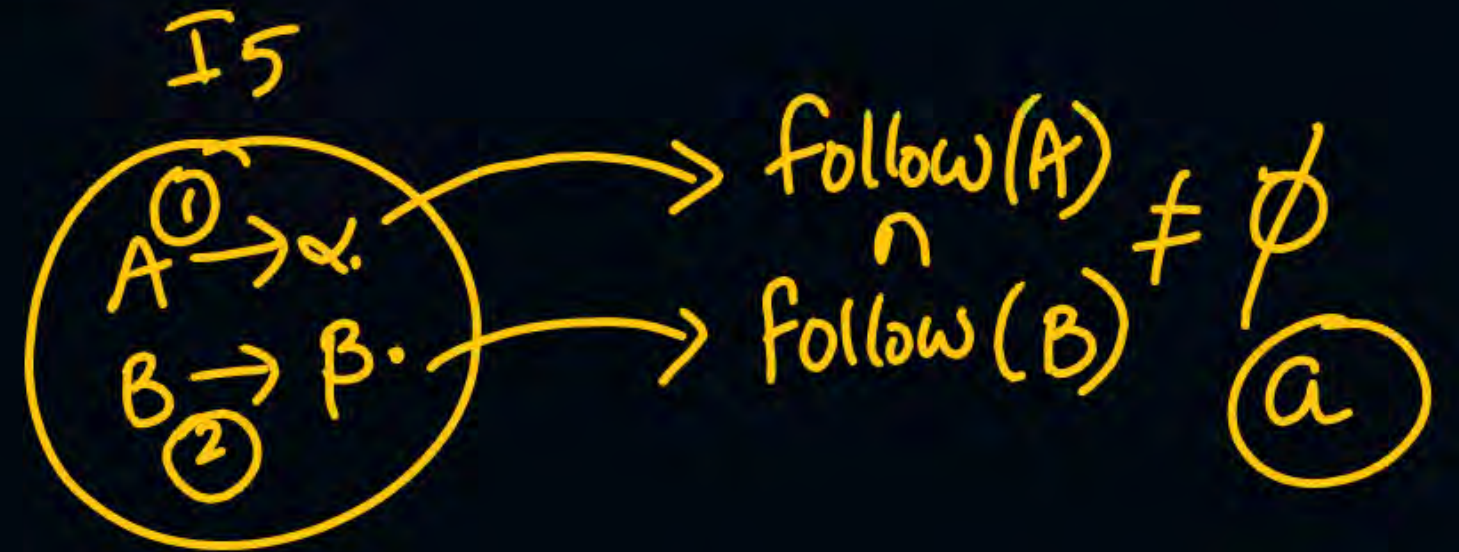
RR Conflict
= ✓

Testing whether a grammar is SLR(1):-



	a
5	S6/6

SR conflict.



	a	b	\$
5	r1/r2		

RR Conflict

LL(1)?
LR(0)?
SLR(1)?

$S \rightarrow dA$
 $\quad \quad /aB$

$A \rightarrow bA$
 $\quad \quad /c$

$B \rightarrow bB$
 $\quad \quad /c$

3 min



THANK - YOU